
From Inverse Optimization to Sequential Decisions: Conformal Prediction for Inverse Reinforcement Learning

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Abstract

1 Inverse reinforcement learning (IRL) is inverse optimization with a sequential
2 forward problem: demonstrations reveal an unknown reward, and the recovered
3 reward is subsequently optimized to prescribe a policy. This perspective exposes
4 the same weakness identified by Conformal Inverse Optimization (CIO) for one-
5 shot decisions: collapsing heterogeneous objectives to one point estimate can
6 yield a bad downstream decision. We introduce **CP-IRL**, which transfers CIO's
7 calibrate-then-robustify principle to Markov decision processes. From a popula-
8 tion of demonstrators, CP-IRL conformally calibrates a set of reward parameters
9 consistent with a new demonstrator and optimizes a policy against the resulting
10 reward ambiguity. The sequential setting requires new machinery. We reduce each
11 calibration score to a reward-dimensional convex program using the Bellman resol-
12 vent, and prove that a fixed transition kernel makes the robust MDP convex for any
13 compact reward ambiguity set; a two-state counterexample shows that this property
14 can fail when ambiguity also enters the dynamics. We establish finite-sample
15 coverage, regret bounds, and a separation example in which point-estimate IRL has
16 unbounded regret while CP-IRL's regret vanishes. Experiments on random MDPs,
17 gridworld, and Objectworld identify the method's scope: robustification reduces
18 true-reward regret under systematic estimation bias, but incurs an insurance cost
19 when the point estimate is merely noisy and approximately unbiased.

20 1 Introduction

21 Inverse reinforcement learning (IRL) is usually introduced as a distinct learning problem: infer a
22 reward from behavior, then optimize that reward to prescribe a policy. Viewed end to end, however,
23 IRL is a sequential instance of *inverse optimization*. The observed decision is a policy or trajectory,
24 the latent objective is a reward function, and the forward problem is an MDP. This connection is more
25 than terminology. It suggests that uncertainty about the recovered objective should be propagated
26 into the downstream decision, rather than discarded after producing a point estimate.

27 Conformal Inverse Optimization (CIO) formalizes exactly this idea for one-shot problems Lin et al.
28 [2024]. CIO replaces a single recovered objective with a conformally calibrated uncertainty set
29 and optimizes robustly over that set. Its guarantee is finite-sample and distribution-free under
30 exchangeability, and its separation example shows why the change matters: point-estimate inverse
31 optimization can retain unbounded downstream regret even with unlimited data. The natural question
32 motivating this paper is therefore:

33 *Can CIO's calibrate-then-robustify principle be transferred from one-shot inverse*
34 *optimization to inverse reinforcement learning without losing coverage or tractabil-*
35 *ity?*

We answer this question with CP-IRL. A population of demonstrators supplies the exchangeable calibration units; each observed policy defines a feasible reward set through Bellman optimality; split conformal prediction calibrates a spherical cap around any chosen IRL point estimator; and a robust MDP converts that calibrated reward uncertainty into a deployed policy. Thus, the conceptual pipeline is inherited from CIO, while the technical work lies in adapting and optimizing it for sequential decisions.

Making the CIO template work for IRL is not a mechanical transplant. Two structural obstacles are specific to the sequential setting:

1. **The nonconformity score is no longer an LP-duality fact.** CIO’s per-point calibration score is computable in closed form because the forward problem is a linear program and reward feasibility is characterized by LP dual feasibility. In an MDP, “this policy is optimal for this reward” is a Bellman-optimality condition, not an LP-duality condition, and computing it naively requires jointly optimizing over the reward parameter *and* the policy’s value function – a variable count that grows with the state space.
2. **The robust decision problem is a minimax game over policies, not points.** CIO’s robust forward problem is a single robust linear program. The MDP analogue is a robust MDP: $\max_{\pi} \min_{\theta}$ over an ambiguity set. General robust MDPs with ambiguity that couples across states are NP-hard Wiesemann et al. [2013]; most tractable results in the literature rely on *rectangularity* assumptions that restrict how the adversary’s choice can vary across states.

We resolve both obstacles and develop the MDP counterpart of CIO’s principal guarantees. The contributions are:

- **Transfer: conformal inverse optimization for IRL.** We formulate one demonstrator as one exchangeable inverse-optimization instance, calibrate a reward uncertainty set through Bellman-optimal reward feasibility, and establish finite-sample coverage together with a true-reward regret bound (§3–§5).
- **Optimization: sequential structure without state-sized calibration** (§4.1). For a fixed demonstrator policy, the induced value function is *linear* in the reward parameter via the Bellman resolvent $(I - \beta P_{\pi})^{-1}$ – the same linear-in-reward value identity underlying the feasible-reward characterization of Ng & Russell Ng and Russell [2000], which we repurpose as a calibration-score reduction; substituting this collapses the per-demonstrator calibration score from a joint optimization over (θ, V) – size $d + |S|$ – to an optimization over θ alone – size d – with no relaxation, by the policy improvement theorem. We further give a Sherman–Morrison–Woodbury update that recomputes this resolvent in $O(k^2|S| + k^3)$ rather than $O(|S|^3)$ when two demonstrators’ policies differ at only k states, motivated by and tailored to the population-of-demonstrators setting.
- **Tractability: a boundary between reward and dynamics ambiguity** (§4.2). We show the robust MDP is a convex program in occupancy-measure space whenever ambiguity is confined to the reward parameter *for any compact ambiguity set*, via a Bellman-recursion-free argument, and – new relative to the closest existing sufficiency-side result (Gadot et al. Gadot et al. [2024], who solve the convex L_p -ball case via Hölder duality and policy gradient) – we prove this is *tight*: an explicit two-state, two-action MDP with a one-dimensional, convex, non-product ambiguity set coupling reward and transition probability breaks concavity, with a hand-verifiable rational-valued Jensen violation.
- **Scope: robustness helps under bias, not noise alone.** Our sequential analogue of CIO’s separation example proves that classical point-estimate IRL can have diverging regret while the calibrated policy’s regret vanishes. Across random MDPs, a region-based gridworld, and Objectworld, we find that calibrated robust hedging reduces true-reward regret specifically when the point estimate is *systematically* biased (as our degenerate construction manufactures), and *increases* regret when the estimate is merely noisy but unbiased – a precise, falsifiable boundary condition for the method’s benefit that we report as a finding, not a limitation to be hidden.

87 2 Related Work

88 **Inverse optimization and its calibration.** Inverse optimization recovers parameters of an optimization problem from observed decisions (Ahuja & Orlin Ahuja and Orlin [2001]; Chan et al. Chan et al. [2018]). Classical formulations return a point estimate; more recent work constructs data-driven uncertainty sets around it (Chenreddy, Bandi & Delage Chenreddy et al. [2022]; Sun, Liu & Li Sun et al. [2023]), but – as Lin, Delage & Chan Lin et al. [2024] show – none of these give the specific guarantee we need: that the *next* observed decision’s true parameter is covered with a stated confidence, distribution-free. Conformal Inverse Optimization is the first to close this gap using split conformal prediction Vovk et al. [2005], and is the paper this work directly extends from one-shot decisions to MDPs.

97 **Feasible reward sets in IRL.** A separate line of work characterizes the *set* of rewards consistent with observed IRL data (Poiani et al. Poiani et al. [2024]; Lazzati & Metelli Lazzati and Metelli [2025]; Skalse et al. Skalse et al. [2022]) and proves PAC-style bounds on recovering it, using it directly as a robust-MDP ambiguity set. This line shares our target object – a feasible reward set characterized by Bellman-optimality constraints – but its guarantees are PAC bounds requiring a generative model of demonstrator sub-optimality; ours require only exchangeability of demonstrators, no assumption on the *process* by which any individual demonstrator deviates from optimality. The transfer question this line raises – whether a feasible reward set identified on one MDP transfers to another (Schlaginhaufen & Kamgarpour Schlaginhaufen and Kamgarpour [2024]) – is a genuinely open problem for our setting that we do not attempt to close here (see §7).

107 **Robust MDPs and the rectangularity boundary.** Robust MDPs with ambiguity coupling across states are NP-hard in general (Wiesemann, Kuhn & Rustem Wiesemann et al. [2013]); tractability is classically recovered via *s*- or *sa*-rectangularity (Iyengar Iyengar [2005]; Nilim & El Ghaoui Nilim and El Ghaoui [2005]), which restricts the adversary to choose independently per state. Recent work extends tractable, *non*-rectangular ambiguity specifically for reward-only uncertainty with a fixed transition kernel: Gadot et al. Gadot et al. [2024] solve the convex L_p -ball case exactly via Hölder duality and a policy-gradient method with proven convergence; Xu et al.’s Group-Robust MDPs Xu et al. [2026] handle *feature-wise* (d-rectangular) ambiguity that couples both reward *and* transition, at the cost of restricting to that rectangular structure. Our sufficiency result (§4.2) covers the same fixed-kernel, reward-only regime as Gadot et al. but for an *arbitrary* compact ambiguity set (not only L_p -balls), via a direct occupancy-measure convex program rather than a policy-gradient method; our necessity result – an explicit demonstration that convexity is lost the moment ambiguity couples into the kernel, even minimally – has no counterpart in either paper.

120 **Conformal prediction in RL-adjacent settings.** Conformal prediction has been used for offline-RL pessimism from Q-value or state-action-coverage uncertainty (UARM, Pan et al. Pan et al. [2026]), for calibrated intervals in infinite-horizon policy evaluation, and for modeling *other agents’ actions* as conformal prediction sets in multi-agent RL (CAMMARL, Gupta & Kahou Gupta and Kahou [2023]) – the last of these is explicitly motivated by an analogy to IRL but never calibrates a reward set. None combine conformal calibration with a reward vector recovered from a population of IRL demonstrators and a robust-MDP regret guarantee, which is the gap this paper closes.

127 **Preference learning and RLHF.** Preference-based and RLHF-style learning fits a reward (or policy) from labeled comparisons or demonstrations, typically from individual expert trajectories (Christiano et al. Christiano et al. [2017]; Rafailov et al. Rafailov et al. [2023]) or crowdsourced preference populations. Our per-demonstrator exchangeable-unit design (§3) is structurally suited to recast “one annotator, one calibration point” for heterogeneous-preference RLHF populations, giving a coverage guarantee (in the intersection-nonemptiness sense of Theorem 2, not parameter containment) of the form “the calibrated reward-ambiguity set is compatible with a γ -fraction of the annotator population’s individual reward functions, which are never directly observed” – a guarantee we are not aware of in the RLHF calibration literature – though we do not run an RLHF-scale experiment here (see §7).

137 3 Setting

138 We now make the CIO–IRL correspondence precise. We adopt the population-of-demonstrators setting used throughout this work: a shared tabular MDP $M = (S, A, P, \beta, \phi)$ with known transition

kernel P , discount β , and feature map $\phi : S \times A \rightarrow \mathbb{R}^d$ inducing reward $r_\theta(s, a) = \phi(s, a)^\top \theta$. A population of demonstrators draws perceived rewards $\hat{\theta}_1, \dots, \hat{\theta}_N \sim P_\theta$ i.i.d. and each acts optimally in M under their own $\hat{\theta}_k$, producing an observed policy $\pi_k = \pi^*(\hat{\theta}_k)$ – we assume the full deterministic policy is observed exactly, not estimated from a finite trajectory sample; relaxing this to trajectory-level observation would make $\hat{\Theta}(\pi_k)$ itself an estimated (rather than exact) feasible set and is outside the scope of the guarantees below.

Assumption 1 (Exchangeable calibration unit). *The pairs $(\hat{\theta}_k, \pi_k)$, $k = 1, \dots, N, N+1$, are i.i.d. (hence exchangeable), and M (in particular P) is shared and fixed across all demonstrators and the deployment instance.*

Under Assumption 1, each demonstrator plays the role of one (decision, context) pair in CIO’s setting, and it is the setting under which CIO’s coverage argument transplants essentially verbatim (§5, Theorem 2). We commit to this “unit A” design over two weaker alternatives – per-task exchangeability with a single demonstrator (still exchangeable, treated as an easy corollary) and per-trajectory exchangeability within one demonstrator (broken by within-trajectory Markov dependence, and used only as an ablation) – and are explicit throughout about what breaks under each relaxation.

The learning task, mirroring CIO’s pipeline, has three steps: (i) split the population into training and calibration sets; (ii) fit a point estimate $\bar{\theta}$ on the training set with any IRL method (we use and compare a sub-optimality-loss linear program, MaxEnt IRL, and Bayesian IRL by MCMC – §6); (iii) calibrate a spherical-cap uncertainty set $C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \theta^\top \bar{\theta} \geq \cos \alpha\}$ on the calibration set and solve the robust MDP $\max_\pi \min_{\theta \in C(\bar{\theta}, \alpha)} \theta^\top \mu(\pi)$ in occupancy-measure space, where $\mu(\pi)$ is π ’s occupancy measure.

4 Method

4.1 Tractable calibration via the Bellman resolvent

CIO’s calibration score for a validation point k is $c_k = \max_\theta \{\theta^\top \bar{\theta} : \theta \in \Theta^{\text{OPT}}(\hat{x}_k, u_k), \|\theta\|_2 \leq 1\}$, where Θ^{OPT} is characterized by LP dual feasibility – tractable because the forward problem is a linear program. In an MDP, the analogous feasible set for a policy π is

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s)) \forall s, \forall a\},$$

the exact set of reward vectors for which π is Bellman-optimal, by the policy improvement theorem. Computing this naively requires the joint optimization $\max_{\theta, V} \theta^\top \bar{\theta}$ subject to the Bellman-optimality inequalities written in terms of both θ and the a priori unknown value function V – a variable count of $d + |S|$.

The key fact we use, which is the sequential-decision restatement of the feasible-reward characterization of Ng & Russell Ng and Russell [2000]: for a **fixed** policy π , the policy’s own induced value function V_θ^π is *linear* in θ , via the Bellman resolvent.

Proposition 1 (Resolvent-based score reduction). *Fix a policy π , let $P_\pi \in \mathbb{R}^{|S| \times |S|}$ be the induced state-transition matrix ($P_\pi(s, s') = P(s, \pi(s), s')$), and let $\Phi_\pi \in \mathbb{R}^{|S| \times d}$ be the feature matrix restricted to the actions π takes ($\Phi_\pi(s, \cdot) = \phi(s, \pi(s))^\top$). Let $W_\pi := (I - \beta P_\pi)^{-1} \Phi_\pi$ (well-defined since $\beta \in [0, 1)$). Then $V_\theta^\pi = W_\pi \theta$ and $Q_\theta^\pi(s, a) = \Psi_\pi(s, a)^\top \theta$ with $\Psi_\pi(s, a) := \phi(s, a) + \beta P(s, a, \cdot)^\top W_\pi$, both linear in θ . Consequently*

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, \Psi_\pi(s, a)^\top \theta \leq \Psi_\pi(s, \pi(s))^\top \theta \forall s, a\},$$

exactly (no relaxation), and the calibration score $c_k = \max_{\theta \in \hat{\Theta}(\pi_k)} \theta^\top \bar{\theta}$ is an optimization over θ alone – size d , not $d + |S|$ – with W_{π_k} (and hence Ψ_{π_k}) precomputed once per distinct policy via a single resolvent solve.

Proof sketch. V_θ^π solves the linear Bellman equation $V_\theta^\pi = \Phi_\pi \theta + \beta P_\pi V_\theta^\pi$, which rearranges to $V_\theta^\pi = (I - \beta P_\pi)^{-1} \Phi_\pi \theta$ since $I - \beta P_\pi$ is invertible for $\beta < 1$. Substituting into $Q_\theta^\pi(s, a) = \phi(s, a)^\top \theta + \beta P(s, a, \cdot)^\top V_\theta^\pi$ gives the stated Ψ_π . The characterization of $\hat{\Theta}(\pi)$ is then the policy improvement theorem’s optimality condition $Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s)) \forall s, a$, rewritten using Ψ_π . $\square$

We verified Proposition 1 against a slow joint- (θ, V) reference implementation to solver precision across randomized instances, and against independent value iteration. The amortized speedup over the joint formulation grows with $|S|$ ($2\text{--}5.5\times$ observed at $|S|$ in the low hundreds). When a new demonstrator’s policy differs from an already-resolved one at only k states, the resolvent updates via a Sherman–Morrison–Woodbury identity in $O(k^2|S| + k^3)$ rather than a fresh $O(|S|^3)$ solve – this incremental update, tailored to the population-of-demonstrators setting, is the part of Proposition 1 without a counterpart in Ng & Russell Ng and Russell [2000].

4.2 A tight tractability boundary for the robust MDP

The robust decision problem, $\max_{\pi} \min_{\theta \in C} \theta^\top \mu(\pi)$, is a minimax game over policies. We work in occupancy-measure space: $M(P)$ denotes the (fixed, since P is known) polytope of achievable occupancy measures μ , normalized so $\sum_{s,a} \mu(s,a) = 1$, and $\Phi \in \mathbb{R}^{|S \times A| \times d}$ is the matrix whose rows are $\phi(s,a)^\top$ (matching the convention already used for Φ_π in §4.1), so that $\theta^\top \Phi^\top \mu = \sum_{s,a} \mu(s,a) r_\theta(s,a)$ is the expected per-step reward of occupancy measure μ under reward θ , and $\theta^\top \Phi^\top \mu / (1 - \beta)$ is the corresponding expected discounted return. The robust objective is $\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$ (the $1/(1 - \beta)$ factor is a positive constant and does not affect the $\arg \max$).

Theorem 1 (Reward-only ambiguity is convex; this is tight). *(a) Sufficiency. For any nonempty compact ambiguity set $C \subseteq \mathbb{R}^d$ (e.g., our calibrated spherical cap $C(\theta, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \theta^\top \theta \geq \cos \alpha\}$, a cap of the unit ball rather than the unit sphere), if the transition kernel P is fixed and known, then $\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$ is a concave maximization over the fixed compact polytope $M(P)$, hence a convex program. (b) Necessity. This can fail as soon as ambiguity couples reward and transition probability, even minimally: there exists a 2-state, 2-action MDP and a convex, one-dimensional, non-product ambiguity set over (θ, P) jointly for which the analogous worst-case value function is not concave.*

Proof sketch. (a) $g_C(x) := \min_{\theta \in C} \theta^\top x$ is a pointwise infimum of affine functions of x , hence concave, regardless of C ’s shape; composing with the linear map $\mu \mapsto \Phi^\top \mu$ preserves concavity; $M(P)$ does not depend on θ , so the ambiguity never touches the feasible set, only the objective, and the outer problem is a concave maximization over a fixed polytope – no Bellman recursion or rectangularity assumption is needed. (b) Take reward $\theta(t) = 1 + t$ and transition probability $q(t) = 1 - t$ coupled through a single scalar $t \in [0, 1]$. The induced worst-case value, as a function of the agent’s (scalar) policy parameter $p \in [0, 1]$, is $W(p) = \min_{t \in [0, 1]} p(1 + t)/(1 - \beta p q(t))$. Direct computation gives the explicit, hand-verifiable, rational-valued Jensen violation $W(5/18) = 10/27 \approx 0.370 < \frac{1}{2}[W(0) + W(5/9)] \approx 0.556$, independently reproduced via numerical optimization; W is therefore not concave, so the outer maximization is not a convex program in this regime. $\square$

Part (a) generalizes the sufficiency result of Gadot et al. Gadot et al. [2024] (who solve the convex L_p -ball case exactly via Hölder duality and policy gradient) to an arbitrary compact C , via a direct occupancy-measure argument. Part (b) shows the coupling required to break concavity is minimal – a single scalar t linking reward and transition ambiguity along a line segment, far short of a fully unconstrained non-rectangular adversary – and, to our knowledge, is not established by either Gadot et al. or Xu et al. Xu et al. [2026]. Note that failure of concavity is a failure of this *particular* tractability argument, not by itself a proof of NP-hardness; general non-rectangular robust MDPs are already known to be NP-hard Wiesemann et al. [2013].

5 Theory

We prove three further results (coverage, regret, separation), each the direct MDP/IRL analogue of a corresponding CIO theorem, and each independently spot-checked computationally against the implemented solvers rather than trusted as derivation alone. A fourth item below (efficiency) is an empirical confirmation, not a theorem, and is reported as such.

233 **Theorem 2** (Coverage). *Under Assumption 1, let $\tau = \lceil (N_{\text{val}} + 1)\gamma \rceil$, let $c_{(\tau)}$ be the τ -th largest*
 234 *calibration score among $c_1, \dots, c_{N_{\text{val}}}$ (Proposition 1), and set $\alpha_\gamma := \arccos(c_{(\tau)})$. Then*

$$\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma,$$

235 *where $\hat{\Theta}_{\text{new}}$ is the new demonstrator’s feasible reward set. **This is an intersection-nonemptiness***
 236 *guarantee, not a containment guarantee: it does not by itself assert $\theta_{\text{new}}^* \in C(\bar{\theta}, \alpha_\gamma)$, only that the*
 237 *calibrated cap is compatible with some reward for which π_{new} is optimal.*

238 The proof is a line-for-line transplant of CIO’s own exchangeability argument, applied to the scores of
 239 Proposition 1 in place of CIO’s LP dual scores: exchangeability of (θ_k, π_k) under Assumption 1 makes
 240 c_{new} exchangeable with the calibration scores, so $c_{\text{new}} \geq c_{(\tau)}$ (equivalently $\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset$)
 241 with probability at least $\tau/(N_{\text{val}} + 1) \geq \gamma$ by the standard rank argument for exchangeable sequences.
 242 A marginal-coverage simulation against the actual calibration code (150 trials per (γ, N_{val}) cell)
 243 confirms coverage $\geq \gamma$ holds at every tested combination; because demonstrator policies are drawn
 244 from a finite set, repeated policies produce tied (atomic) calibration scores, which is the most likely
 245 source of the over-coverage we observe empirically beyond the $1/(N_{\text{val}} + 1)$ slack the rank argument
 246 already permits (§6).

247 **Theorem 3** (Regret bound). *Let $\|\phi\|_{2,\infty} := \max_{s,a} \|\phi(s, a)\|_2$, $\nu(\mu) := \|\Phi^\top \mu\|_2 / (1 - \beta) \leq \bar{\nu} :=$
 248 $\|\phi\|_{2,\infty} / (1 - \beta)$ for any occupancy measure $\mu \in M(P)$ (a Lipschitz constant for $\theta \mapsto \theta^\top \Phi^\top \mu / (1 - \beta)$)
 249 in $\bar{\theta}$, via Cauchy–Schwarz), and let μ_{CIRL} , μ^* denote the occupancy measures of the robust policy and
 250 of the true optimum $\pi^*(\theta_{\text{new}}^*)$, respectively. Suppose the calibrated cap $C(\bar{\theta}, \alpha_1)$ satisfies the almost-
 251 sure-covering condition $\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_1) \neq \emptyset) = 1$ for a fresh demonstrator (strictly stronger
 252 than Theorem 2’s γ -coverage; achievable trivially by taking C to be the whole unit ball, i.e. α_1 at its
 253 maximal value, but then uninformative), that any two unit-norm rewards optimal for the same policy
 254 are within η of each other (bounded feasible-set diameter), and that the population mean reward is
 255 within σ of θ_{new}^* in ℓ_2 . Then the **average optimality gap** $\text{AOG} := V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}})$ (the
 256 true-reward regret of the robust policy relative to the in-hindsight optimum) satisfies*

$$\text{AOG} \leq (2 \sin \alpha_1 + \eta + \sigma) \nu(\mu^*) + (\eta + \sigma) \nu(\mu_{\text{CIRL}}),$$

257 *and substituting the uniform bound $\bar{\nu}$ for both $\nu(\mu^*)$ and $\nu(\mu_{\text{CIRL}})$ gives the fully distribution-free*
 258 *corollary $\text{AOG} \leq (2 \sin \alpha_1 + 2\eta + 2\sigma) \bar{\nu}$.*

259 The proof is a three-term telescoping argument (Lipschitz continuity of $\theta \mapsto \theta^\top \Phi^\top \mu$, the chord-
 260 length identity $\|\theta - \theta'\|_2 = 2 \sin(\psi/2)$ for unit vectors at angle ψ – giving $2 \sin \alpha_1$ as the chord
 261 bound for any two points within $C(\bar{\theta}, \alpha_1)$, since they are within angle $2\alpha_1$ of each other – and
 262 the robust policy’s minimax optimality), re-derived from scratch for the MDP’s $\max_\pi \min_\theta$ sign
 263 convention rather than transplanted line-for-line from CIO’s $\min_x \max_\theta$ convention. We verified
 264 $\bar{\nu}$ as a valid uniform Lipschitz constant against 200 randomized MDPs with zero violations, and
 265 separately verified the almost-sure-covering hypothesis and the bound’s numeric validity by direct
 266 computation on random instances. We are explicit that this is a general worst-case bound, not a
 267 tight one: cross-checked against the exact closed-form regret derived for our degenerate construction
 268 below (where the almost-sure-covering condition holds already at $\alpha_1 = 0$, a special feature of that
 269 construction), the general bound’s looseness grows with the construction’s scaling parameter – from
 270 roughly $8\text{--}15\times$ at the low end of the range tested to roughly $390\text{--}430\times$ at the high end, because the
 271 exact regret there decays to zero while $\eta, \sigma, \bar{\nu}$ converge to nonzero constants – a qualitatively worse
 272 (growing, not fixed-order) gap than CIO’s own Example 1 exhibits for its own general bound (roughly
 273 $2.3\text{--}4\times$, staying bounded because CIO’s construction keeps a richer feasible region that does not
 274 saturate the same way).

275 **Theorem 4** (Separation). *There exists a single-state, two-action MDP (the sequential-decision*
 276 *analogue of CIO’s degenerate Example 1, with reward features the negation of CIO’s original cost*
 277 *features) and an IRL point estimator such that: the point-estimate solution converges to a degenerate*
 278 *direction $\theta_u = (1, u)/\sqrt{1+u^2} \neq \theta^*$ (a unit vector, hence on the boundary of the unit ball that*
 279 *$C(\bar{\theta}, \alpha)$ is a cap of) as the scaling parameter $u \rightarrow \infty$, giving classical point-estimate IRL true-reward*
 280 *regret that grows without bound in u ; while the calibrated robust policy’s regret is bounded and*
 281 *shrinks to zero over the same range. The robust optimum is a genuine mixture, with exact closed form*
 282 *$p^*(u) = u^2/(1+u^2)$, independent of the calibration angle.*

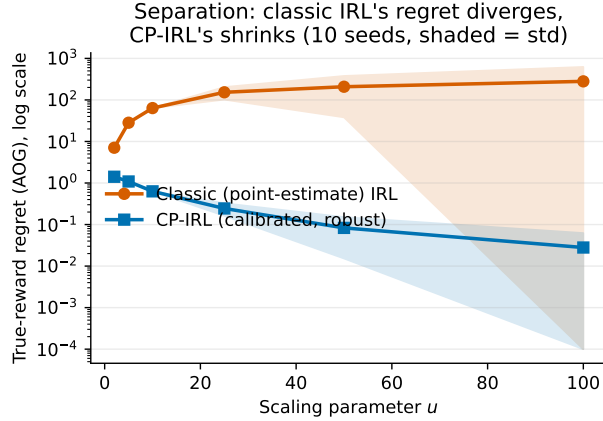


Figure 1: Separation result (Theorem 4): true-reward regret of the deployed policy, measured against the ground-truth reward θ_{new}^* , vs. the scaling parameter $u \in [2, 100]$ (log scale), 10 seeds, shaded regions show ± 1 std. “Classic IRL” is the sub-optimality-loss linear-program point estimator (§6), which converges to the degenerate direction θ_u ; its regret (orange) grows monotonically and without bound. CP-IRL’s regret (blue) shrinks monotonically toward zero over the same range. Variance grows with u for both curves, consistent with tie-breaking sensitivity as $\bar{\theta}$ approaches the exact degenerate boundary.

283 Theorem 4 is proved via Sion’s minimax theorem for the mixture’s closed form, and independently
 284 re-verified against the actual implemented solver to 10^{-6} precision. Figure 1 shows the live 10-
 285 seed computational confirmation. We flag directly that this separation exhibits the LP-based point
 286 estimator’s failure mode specifically (a degenerate boundary direction under tie-breaking); we discuss
 287 whether it extends to MaxEnt and Bayesian IRL in §6, Q3.

288 **Efficiency.** We confirm, empirically, that the calibrated set is not vacuously wide: with a *unified* noise
 289 parameter controlling all sources of demonstrator population variation simultaneously, the calibrated
 290 angle shrinks monotonically from 27.7° to effectively 0° as demonstrator-population noise vanishes.

291 6 Experiments

292 Our experiments answer three questions: (Q1) does calibration attain its target coverage without
 293 becoming vacuous; (Q2) when does robustification improve downstream regret; and (Q3) are the
 294 conclusions tied to one IRL estimator or environment? We evaluate three point estimators – a sub-
 295 optimality-loss linear program, MaxEnt IRL, and Bayesian IRL (Metropolis–Hastings posterior mean
 296 under a Boltzmann-rational demonstrator likelihood) – each combined with three decision rules – the
 297 plain point estimate, our calibrated robust policy, and a “concentration baseline” robust policy whose
 298 uncertainty-set radius comes from a classical Dvoretzky–Kiefer–Wolfowitz concentration bound
 299 rather than the exact split-conformal quantile – across three environments: uncorrelated random
 300 tabular MDPs, a region-based gridworld with spatially contiguous (Voronoi) reward features, and
 301 Objectworld, the standard IRL benchmark with smooth, continuous distance-to-object features. All
 302 experiments run on tabular MDPs at a scale ($|S| \leq 36$, $d \leq 5$) chosen to fit within a laptop-class
 303 compute budget (20 CPU cores, no GPU dependency, no commercial solver).

304 **Q1: Coverage and efficiency.** Realized coverage tracks the target γ across every environment and
 305 estimator tested (e.g. ~ 0.88 – 0.89 at $\gamma = 0.8$ on the gridworld), and the calibrated angle responds
 306 monotonically and sensibly to γ , to demonstrator noise, and to the train/calibration split ratio. The
 307 DKW concentration baseline is consistently and measurably looser than exact conformal calibration
 308 at matched nominal confidence, across every estimator and every environment tested.

309 **Q2: The benefit depends on systematic bias.** On our degenerate construction (§5), robust hedging
 310 dramatically reduces true-reward regret, because the point estimate there is *systematically* miscali-
 311 brated by construction. On all three generic environments tested – random MDPs, the gridworld, and
 312 Objectworld – under realistic demonstrator-population noise, robust hedging *increases* true-reward

regret relative to the plain point estimate, consistently across all three point estimators, holding across a full sweep of the coverage target γ and of the train/calibration split ratio. We verified this is not an implementation defect three independent ways: the robust solver’s reported worst-case value matches an independent second-order-cone-program oracle to solver precision; the robust decision’s worst-case value is confirmably *better* than the plain estimate’s worst-case value over the same calibrated set; and we directly measured that these environments’ optimal policies are sensitive to reward direction – a 14° reward perturbation already changes the optimal action at roughly 10–13% of states across the three environments tested (9.6% Objectworld, 9.9% gridworld, 13.1% random MDPs – a modest, not dramatic, margin). Critically, Objectworld’s lower policy-sensitivity (measured directly, and lower than both the gridworld and random MDPs) did **not** flip the result: robust hedging still costs true-reward regret there, ruling out landscape smoothness as the deciding factor and leaving systematic bias in the point estimate as the consistent, mechanistically understood explanation across all three environments.

Q3: The conclusion persists across estimators and environments. The gridworld comparison combines all three point estimators with the plain, conformal-robust, and DKW-robust decision rules. MaxEnt and Bayesian IRL recover an accurate point direction, whereas the LP estimator has nontrivial error; in all three cases, conformal robustification is less conservative than DKW but incurs higher true-reward regret than the corresponding point policy. The same direction holds in Objectworld. We have not run MaxEnt or Bayesian IRL on the degenerate construction of Theorem 4 itself – that theorem’s divergence result is specific to the LP estimator’s tie-breaking behavior at the exact optimality boundary, and we do not claim MaxEnt or Bayesian IRL (which are strictly interior, likelihood-based estimators) necessarily diverge there too; whether the separation result generalizes to these smoother estimators is left open. These comparisons separate two claims that are often conflated: conformal calibration can produce a tighter valid uncertainty set than a generic concentration bound, yet optimizing for its worst case need not improve average performance under the true reward.

7 Discussion and Limitations

When does CP-IRL help, precisely? Our theory (Theorem 4) proves a separation on a construction where the point estimate is *systematically* biased – classical point-estimate IRL’s regret diverges there while CP-IRL’s vanishes. Our experiments show that on generic environments with unbiased-but-noisy point estimates, it does not, and in fact costs regret in exchange for a worst-case guarantee that is real (independently verified) but not free. This is a precise, falsifiable scope condition, not a hedge: the method should be deployed when there is reason to believe point estimation is subject to a genuine structural bias rather than applied by default whenever demonstration data is merely noisy. Characterizing which real-world settings exhibit this kind of systematic bias, and whether it can be diagnosed from data without knowing the ground truth, is an open problem this paper does not resolve.

What is not yet established. The general-case regret bound is provably loose – up to several hundred-fold on the one instance where we can compare it against an exact result – and should be read as a worst-case guarantee, not a predictor of typical-case performance. A source-to-target probe further shows that a reward set calibrated on one MDP does *not* safely transfer to a related MDP with different dynamics: at target coverage 0.8, changing the gridworld slip probability from 0.10 to 0.15 lowers mean target coverage to 0.567 over eight seeds. The probe establishes the failure of naive transfer, not its remedy. Developing a weighted-conformal correction that combines covariate-shift weighting (in the sense of Tibshirani et al. [2019]) with the principal-angle transferability conditions of Schlaginhaufen & Kamgarpour [2024] remains future work. We also tested a preference-population/RLHF-style relabeling of our per-demonstrator design: a “response selection” MDP in which the population is reinterpreted as RLHF annotators rather than task demonstrators, using the same calibration and robust-MDP code unchanged. Realized coverage meets or exceeds the target γ at every level tested (0.947–0.994 across $\gamma \in \{0.5, 0.7, 0.8, 0.9\}$, 8 seeds), consistent with Theorem 2 and confirming the framework generalizes by relabeling alone; the gap above target is likely the same score-atom over-coverage discussed under Theorem 2, sharpened here by a small, low-dimensional annotator population, and is not a special exemption for the RLHF framing (the same systematic-bias-conditional benefit applies). All experiments are at tabular scale by

367 explicit design choice; we make no claim about function-approximation or deep-RL/language-model
368 settings.

369 8 Conclusion

370 IRL is inverse optimization with an MDP as its forward problem. Starting from this connection,
371 we transfer CIO’s calibrate-then-robustify principle to reward learning and supply the sequential
372 machinery the transfer requires: a resolvent-based calibration score, a convex robust-MDP formulation
373 for compact reward ambiguity, and coverage, regret, and separation guarantees. The same connection
374 also clarifies the method’s limits. Calibration certifies uncertainty; it does not make worst-case
375 optimization free. In our experiments, the resulting policy improves true-reward regret when point
376 estimation is systematically biased, but pays an insurance cost when estimation is merely noisy. This
377 distinction is the practical conclusion of CP-IRL and the main condition governing its use.

378 References

- 379 Ravindra K. Ahuja and James B. Orlin. Inverse optimization. *Operations Research*, 49(5):771–783,
380 2001. doi: 10.1287/opre.49.5.771.10607.
- 381 Timothy C. Y. Chan, Taewoo Lee, and Daria Terekhov. Inverse optimization: Closed-form solutions,
382 geometry, and goodness of fit. *Management Science*, 65(3):1115–1135, 2018. doi: 10.1287/mnsc.
383 2017.2992.
- 384 Abhilash Reddy Chenreddy, Nymisha Bandi, and Erick Delage. Data-driven conditional robust
385 optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- 386 Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep
387 reinforcement learning from human preferences. In *Advances in Neural Information Processing
388 Systems (NeurIPS)*, pages 4299–4307, 2017.
- 389 Uri Gadot, Esther Derman, Navdeep Kumar, Maxence Elfatihi, Kfir Levy, and Shie Mannor. Solving
390 non-rectangular reward-robust mdps via frequency regularization. In *Proceedings of the AAAI
391 Conference on Artificial Intelligence*, volume 38, pages 21090–21098, 2024.
- 392 Nikunj Gupta and Samira Ebrahimi Kahou. Cammarl: Conformal action modeling in multi agent
393 reinforcement learning. *arXiv preprint arXiv:2306.11128*, 2023.
- 394 Garud N. Iyengar. Robust dynamic programming. *Mathematics of Operations Research*, 30(2):
395 257–280, 2005. doi: 10.1287/moor.1040.0129.
- 396 Filippo Lazzati and Alberto Maria Metelli. Robustness in the face of partial identifiability in reward
397 learning. *arXiv preprint arXiv:2501.06376*, 2025.
- 398 Bo Lin, Erick Delage, and Timothy C. Y. Chan. Conformal inverse optimization. In *Advances in
399 Neural Information Processing Systems (NeurIPS)*, 2024.
- 400 Andrew Y. Ng and Stuart J. Russell. Algorithms for inverse reinforcement learning. In *Proceedings
401 of the Seventeenth International Conference on Machine Learning (ICML)*, pages 663–670, 2000.
- 402 Arnab Nilim and Laurent El Ghaoui. Robust control of markov decision processes with uncertain
403 transition matrices. *Operations Research*, 53(5):780–798, 2005.
- 404 Licheng Pan, Haocheng Yang, Haoxuan Li, Yichen Sun, Yunsheng Lu, Shijian Wang, Lei Shen,
405 Yuan Lu, Zhixuan Chu, and Hao Wang. Uncertainty-aware reward modeling for stable rlhf. *arXiv
406 preprint arXiv:2606.19818*, 2026.
- 407 Riccardo Poiani, Gabriele Curti, Alberto Maria Metelli, and Marcello Restelli. Sub-optimal ex-
408 perts mitigate ambiguity in inverse reinforcement learning. In *Advances in Neural Information
409 Processing Systems (NeurIPS)*, pages 85778–85823, 2024.
- 410 Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea
411 Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances
412 in Neural Information Processing Systems (NeurIPS)*, pages 53728–53741, 2023.

- 413 Andreas Schlaginhaufen and Maryam Kamgarpour. Towards the transferability of rewards recovered
414 via regularized inverse reinforcement learning. In *Advances in Neural Information Processing*
415 *Systems (NeurIPS)*, pages 21461–21501, 2024.
- 416 Joar Skalse, Matthew Farrugia-Roberts, Stuart Russell, Alessandro Abate, and Adam Gleave. In-
417 variance in policy optimisation and partial identifiability in reward learning. *arXiv preprint*
418 *arXiv:2203.07475*, 2022. Also published at ICML 2023, PMLR vol. 202, pp. 32033–32058.
- 419 Chunlin Sun, Linyu Liu, and Xin Li. Predict-then-calibrate: A new perspective of robust contextual
420 lp. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- 421 Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal
422 prediction under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*,
423 pages 2526–2536, 2019.
- 424 Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random*
425 *World*. Springer, 2005. ISBN 978-0387001524.
- 426 Wolfram Wiesemann, Daniel Kuhn, and Berç Rustem. Robust markov decision processes. *Mathe-*
427 *matics of Operations Research*, 38(1):153–183, 2013. doi: 10.1287/moor.1120.0566.
- 428 Mingyuan Xu, Zongqi Xia, Tianxi Cai, Doudou Zhou, and Nian Si. Learning sequential de-
429 cisions from multiple sources via group-robust markov decision processes. *arXiv preprint*
430 *arXiv:2602.01825*, 2026.